

Conjunction, Disjunction and Negation

Let's look at a few examples:

Which of the following are mathematical statements? If it is a statement, then determine its Truth Value if possible:

- 1. I have magical powers and I am not a human**
- 2. $1 = 0$**
- 3. I have 3 apples in my hand or I am allergic to apples**
- 4. $1+3$**
- 5. $2+2 = 4$ or $3+1 = 4$**
- 6. $4 \in \mathbf{R}$ and $4 < 3^2$**

Let's look at each one of the claims :

- 1. I have magical powers and I am not a human**

- How do you determine if I am telling the truth or I am lying?

Well, if I am telling the truth, I need to have magical powers **and** I need to be not human. If even one of the two claims is false, I am lying. So, the above claim will have only one truth value, hence it is indeed a statement.

- 2. $1 = 0$**

- We know that the above is a statement and the truth value of the above statement is False.

- 3. I have 3 apples in my hand or I am allergic to apples.**

- Can we determine the truth value of the above sentence?

Well, in order for the above sentence to be true, I just need one of the claims to be true. If I have 3 apples in my hand, I don't need to check if I am allergic to apples; the above sentence is True and I am telling the truth.

Similarly, if I do not have 3 apples in my hand but I am allergic to apples, I will still be telling the truth and the above sentence will be True. I only way you can say that I am lying is when I do not have 3 apples in my hand and I am not allergic to apples. In all the cases, there is no ambiguity about the truth value of the sentence, hence it is a mathematical statement.

4. $1 + 3$

- The above is just a phrase the truth value of which cannot be determined.

Hence it is **not** a statement

Try 5 and 6 for yourself using the above examples!

In some of the above examples, we saw that the statements were built from simpler statements using “And/Or”. They are known as **logical connectives**.

The logical Connectives of sentential logic are:

And/Conjunction

Or/Disjunction

Negation/ Not

Conditional/If-then/Implication

Biconditional/If and only if

In this chapter we will cover the first three and continue with the last two in the following chapter.

Conjunction / And

If we have two statements: P, Q → We represent statements using upper case letters eg: $P(x)$
To say that P and Q is true, both P needs to be True and Q needs to be True.

Even if one of the statements is False, “ P and Q ” is False.

We use the symbol “ $\wedge$ ” to represent Conjunction/ And. This is called conjunction of P and Q . → $P \wedge Q$ $P \wedge Q \wedge R$

Let's Practice:

What is the Truth Value of the following Statements:

1. $\frac{\text{T}}{(2+3=5)} \wedge \frac{\text{T}}{(\forall x \in \mathbb{R} . x^2 \geq 0)}$ True
2. $\frac{\text{F}}{(1+4=3)} \wedge (\forall x \in \mathbb{R} . x^2 \geq 0)$ False
3. $\frac{\text{F}}{(1+3=5)} \wedge (\exists x \in \mathbb{R} . x^2 < 0)$ False

Disjunction/ OR

If we have two statements: P, Q

To say that P or Q is True means that either P is True or Q is True (or both are True). So, if one of the statements is true, the entire statement has a Truth value of True. If both the statements are False, then “P or Q” is False.

We use the symbol “ $\vee$ ” to represent Disjunction/ Or. This is called disjunction of P and Q. $\rightarrow P \vee Q$

Let's Practice:

What is the truth value of the following statements:

1. $\overbrace{(2+3=5)}^{\text{True}} \vee \underbrace{(\forall x \in \mathbb{R} . x^2 \geq 0)}_{\text{True}} \quad \text{True}$
2. $\underbrace{(1+4=3)}_{\text{False}} \vee \underbrace{(\forall x \in \mathbb{R} . x^2 \geq 0)}_{\text{True}} \quad \text{True}$
3. $\underbrace{(1+3=5)}_{\text{False}} \vee \underbrace{(\exists x \in \mathbb{R} . x^2 < 0)}_{\text{False}} \quad \text{False}$

Negation/ NOT

If we have a statement P,

To say that Not P is True means that P is False. So when P is True, Not P is False and when P is False, Not P is True.

We use the symbol “ $\neg$ ” to represent Negation/Not. This is called the Negation of P.

$\rightarrow \neg P$

In general, the negation of a universal (i.e. “ $\forall$ ”) claim is one of existence (i.e. “ $\exists$ ”), and vice versa.

How?

Let’s look at an example:

Suppose we have a set A and $P(x)$ is a claim about x and we have the following statement:

$$\forall x \in A. P(x)$$

The above statement says that no matter which element , x , from A we choose, the claim $P(x)$ will be True for all such elements. So, how do we negate this statement/ prove that this statement is False?

If we could find at least one element, x , from set A for which the claim $P(x)$ is False, we have proven that the above statement is False.

So, if we were to write this mathematically using the notations we have learnt till now, it would be:

$$\exists x \in S \text{ such that } P(x) \text{ is False}$$

We know now that notational “ $\neg$ ” is used to represent “logical negation” or “not”.

$$\neg(\exists x \in A. x^2 < 1) \\ \rightarrow \forall x \in A. x^2 \not< 1$$

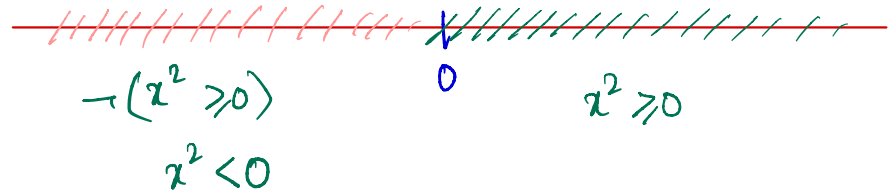
So, the above negated statement can be written as:

$$\forall x \in A. x^2 \geq 1$$

$$\neg(\forall x \in A. P(x))$$

Or

$$\exists x \in A. \neg P(x)$$



Let's make it more concrete:

Negate the following statement:

$$\forall x \in \mathbb{R}. x^2 \geq 0$$

$$\neg(\forall x \in \mathbb{R}. x^2 \geq 0) \\ \neg(\forall x \in \mathbb{R}). \neg(x^2 \geq 0) \\ \underline{\exists x \in \mathbb{R}. x^2 < 0}$$

The above statement can be read as:

For all real numbers, x , the square of the number is always greater than/equal to 0.

How do we make this statement False?

We need at least one such element, x , belonging to the set of Real numbers the square of which is not greater than/ equal to 0

OR

We need at least one such element, x , belonging to the set of Real numbers the square of which is less than 0.

Can we say $\neg (\forall x \in S. P(x))$
is $\forall x \notin S. \neg P(x)$

Now, if we were to write this in terms of Mathematical notations, here's how we can do it:

$$\exists x \in \mathbb{R}. x^2 < 0$$

This is used for defining a “**Counterexample**” to a given statement!

Let's Practice:

Negate the following statements:

1. $\forall x \in \mathbb{R}. y = x^3$ $\neg (\forall x \in \mathbb{R}. y = x^3) \Rightarrow \exists x \in \mathbb{R}. y \neq x^3$

2. $\forall x \in \mathbb{R}. \neg P(x)$, where $P(x)$ is given by: $\forall y \in \mathbb{R}. x = y^3$

3. $\forall a \in A. \exists b \in B. \forall c \in C. \exists d, e \in D. P(a, b, c, d, e)$ ^[LSEP]
 $\exists a \in A. \forall b \in B. \exists c \in C. \forall d, e \in D. \neg (P(a, b, c, d, e))$

$\neg (\forall x \in \mathbb{R}. \neg P(x))$
 $\neg (\forall x \in \mathbb{R}. \neg (\forall y \in \mathbb{R}. x = y^3))$
 $\exists x \in \mathbb{R}. \neg (\neg (\forall y \in \mathbb{R}. x = y^3))$
 $\exists x \in \mathbb{R}. \forall y \in \mathbb{R}. x = y^3$